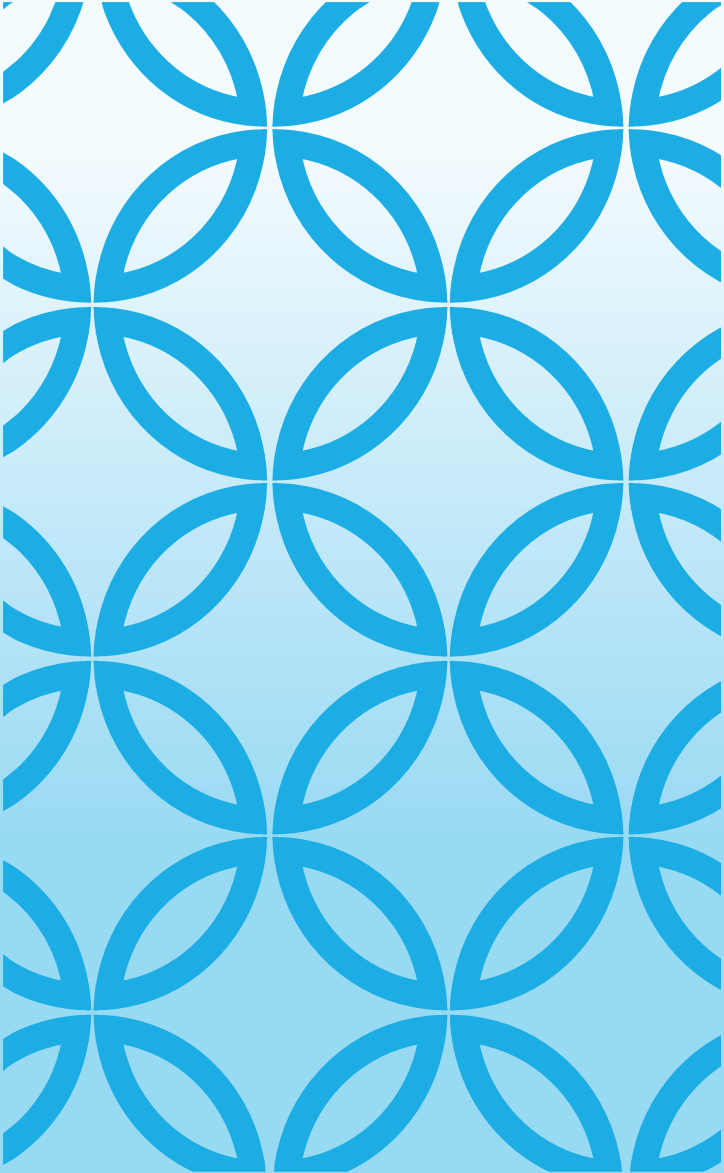




SLIME MOULD OPTIMIZATION ALGORITHMS

Urban Planning & Space Exploration Pathfinding

Presented By Suhaan Balpande, Ranvir Bhakri, Parth Asapuri, Ayush Baluapuri

BIOLOGICAL PHENOMENON

Slime moulds are brainless, single-celled but multinucleate organisms that behave like intelligent systems forming networks. They solve optimization problems, and adapt to their environment. They are neither a fungi nor animal but belong to diverse protist group, called *Physarum Polycephalum* being the most studied species



Source: [US National Park Photo](#)

TYPES OF SLIME MOULD

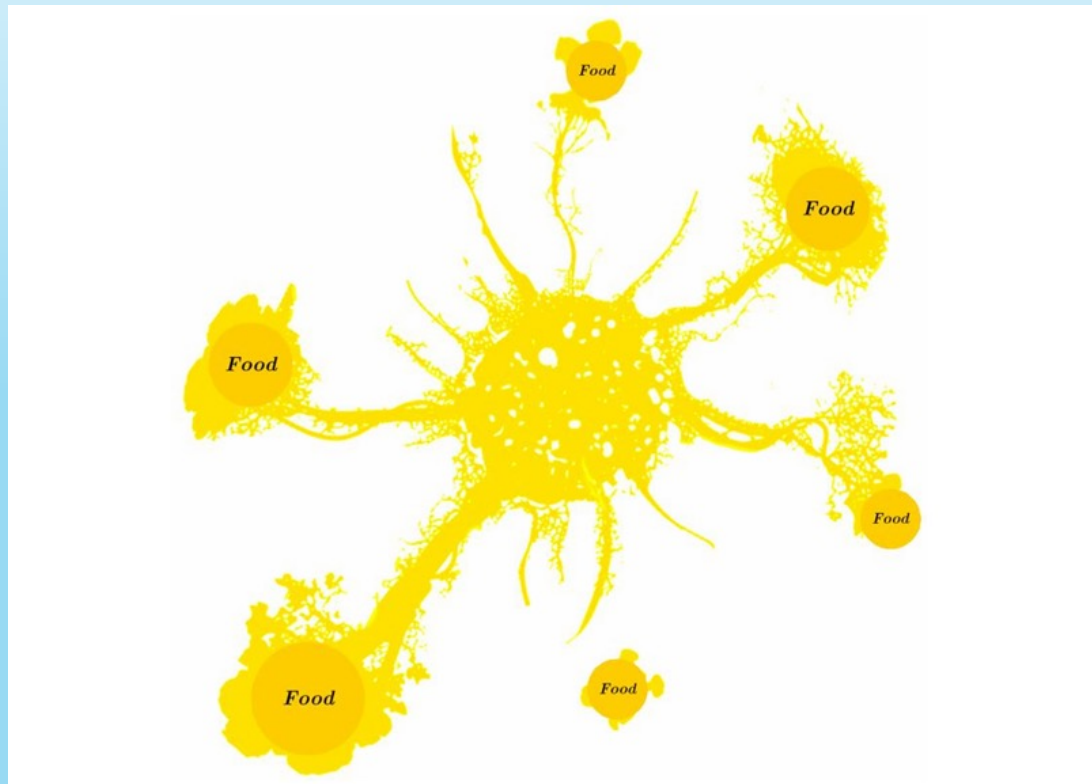
1. Plasmodial Slime Moulds (Myxomycetes)
2. Cellular Slime Moulds (Dictyostelids)
3. Protostelid Slime Moulds
4. Plasmodiophorid Slime Moulds (Parasitic)



Source: [Google](#)

WHICH ONE THE ALGORITHM IS BASED ON?

The yellow slime mould we will be talking about is *Physarum polycephalum*



PHYSARUM POLYCEPHALUM

Biological Traits:

Cytoplasmic streaming

Oscillatory contractions

Adaptive optimization

CYTOPLASMIC STREAMING

Slime moulds transport nutrients and chemical signals through rhythmic pulses of cytoplasm that flow inside their vein like networks. When food is abundant, these flows intensify, causing veins to thicken and strengthen in that direction. This simple mechanism allows the organism to allocate resources efficiently and build transport pathways that resemble optimized networks.



OSCILLATORY CONTRACTIONS

The veins of slime moulds contract and relax in cycles, acting like a biological oscillator. When food is detected, positive feedback strengthens the contractions along that path, reinforcing growth toward the source. Conversely, negative feedback weakens contractions in less promising directions, causing those veins to shrink. This push pull system guides the organism adaptively, ensuring energy is invested where it yields the greatest benefit.

Now Lets see an video in which we can observe these contractions.



Source: [Google GIF](#)

ADAPTIVE OPTIMIZATION

No brain, no nerves → yet capable of solving complex spatial problems.

Exploration strategy → spreads widely to sample multiple directions.

Exploitation strategy → focuses growth on the richest food sources.

Near-optimal solutions → finds shortest paths, efficient transport networks, and resilient connections.

SLIME MOULD REPRODUCTION

Isogamous gametes → all reproductive cells are the same size (no egg/sperm distinction).

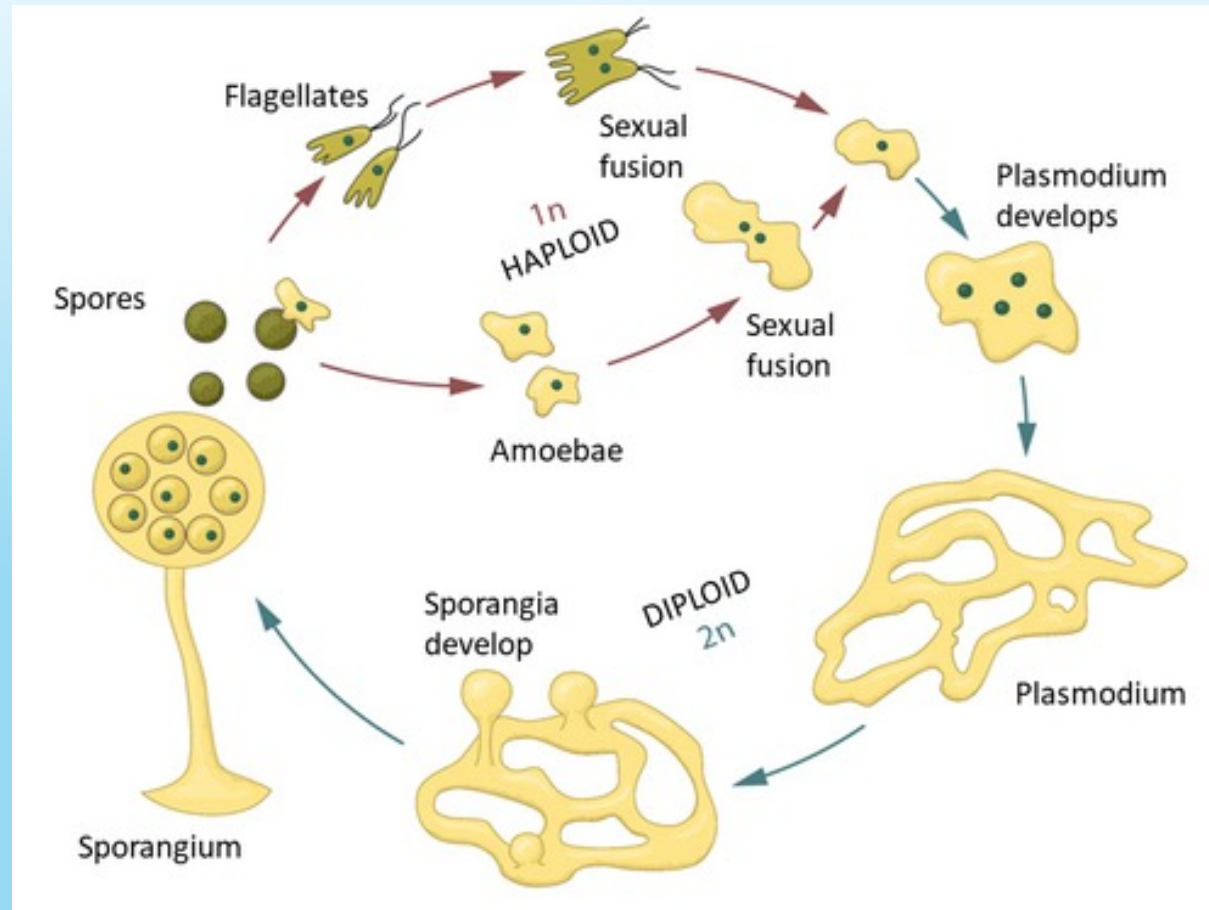
Genetic system → three genes (matA, matB, matC) with multiple variants.

Diploid stage → each mature slime mould carries two copies of each gene.

Gamete diversity → up to 8 cell types per individual; across populations, >500 mating types possible.

Biological advantage → increases chance of finding partners and reduces inbreeding risk.

SLIME MOULD LIFE CYCLE



EXTRA VIDEO ABOUT SLIME MOULDS



Source: [Seeker YT](#)

PROBLEM STATEMENT

The core problem statements addressed by the Slime Mould Algorithm(SMA) focus on overcoming optimization challenges like local optima traps, poor population diversity, and slow convergence when solving complex mathematical, engineering, and pathfinding tasks

The Slime Mould Algorithm is designed to efficiently solve complex optimization problems by finding the best possible solution in a large search space while avoiding local optima through nature inspired search behavior.

The Slime Mould Algorithm (SMA), introduced in 2020, is inspired by the food searching behavior of the slime mould *Physarum polycephalum*. It imitates how slime mould adapts its movement and network structure to locate the richest food source, enabling the algorithm to search efficiently for the best solution while avoiding poor solutions.

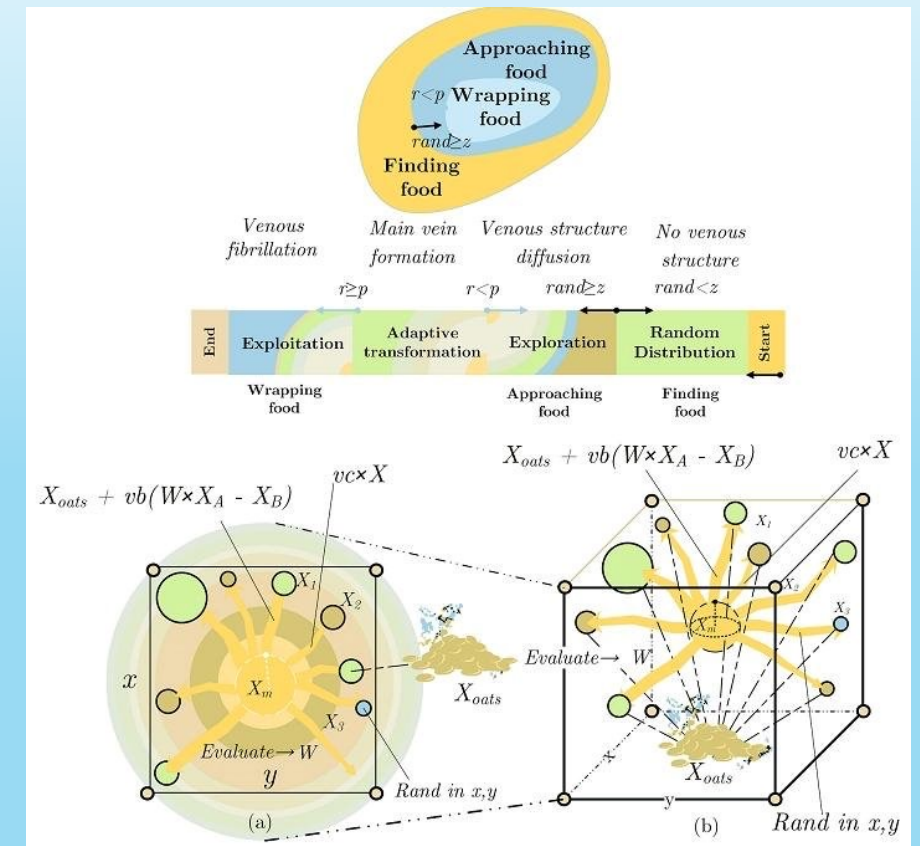
BIOLOGICAL BEHAVIORS OF SMA

Approaching Food: Simulates the shifting of position based on the concentration and quality of the objective function.

Wrapping Food: Mimics the contraction and adjustment of venous thickness based on cytoplasmic streaming velocity.

Grapple Food (Oscillation): Uses stochastic parameters to mimic biological waves that force the population to explore alternative regions or lock onto optimal parameters.

The Slime Mould Algorithm (SMA) addresses this problem by mimicking the food-searching, adaptive movement, and network-forming behavior of slime mould. It uses these biological behaviors to balance exploration and exploitation and search for an optimal solution.



CHALLENGES IN OPTIMIZATION AND FOR SMA

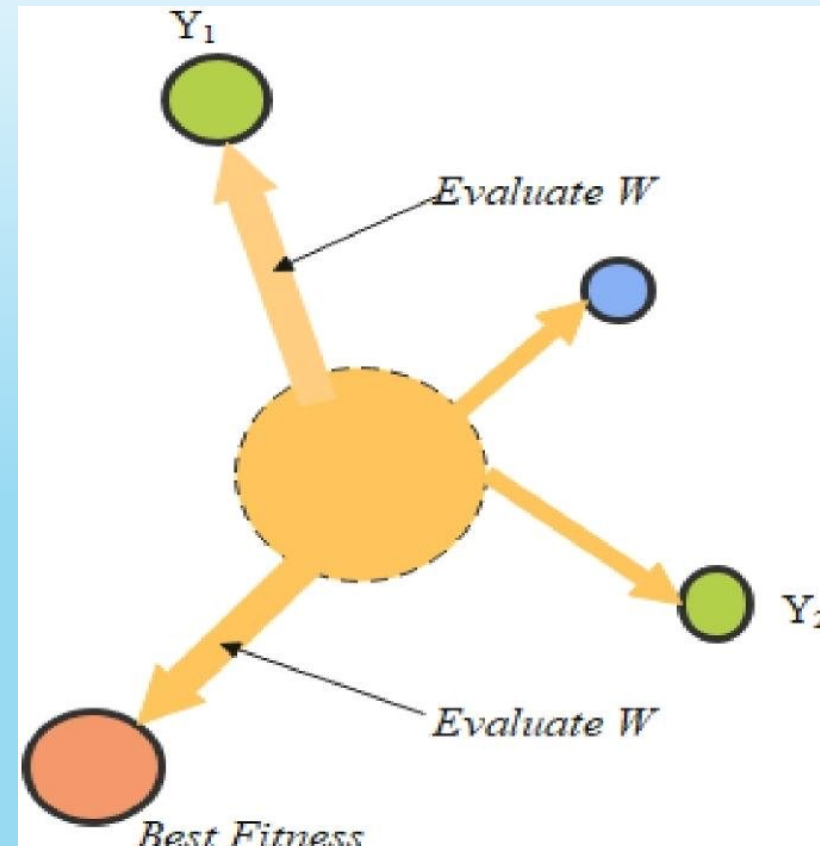
The optimization problems can be difficult because the search space may be complex, nonlinear, and contain many local optimum solutions.

Traditional optimization methods may become slow or get trapped in a poor solution.

Therefore, there is a need for an efficient optimization technique that can explore different possible solutions and identify a near-optimal solution.

The Slime Mould Algorithm addresses this problem by mimicking the intelligent foraging and adaptive.

Many real-world optimization problems involve a large and complex search space, making it difficult to find the best solution using traditional optimization techniques.



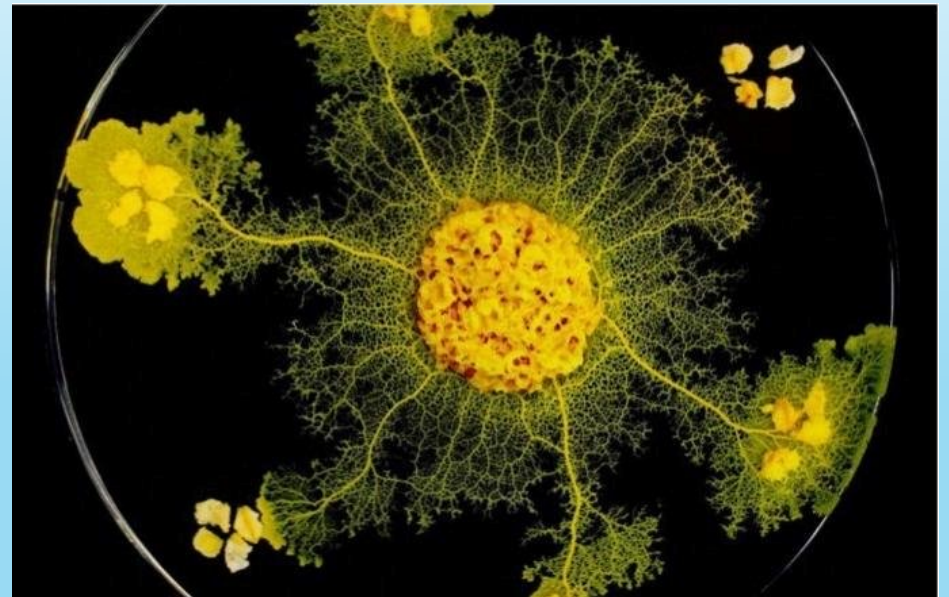
PROBLEM SOLVING OF SMA

The problems may contain multiple possible solutions, nonlinear relationships, and local optimum points where an algorithm can become trapped.

Therefore, there is a need for an efficient optimization technique that can explore a large search space, avoid local optima, and find a high-quality or near-optimal solution within a reasonable amount of time.

It addresses this challenge by mathematically modelling the adaptive food-searching behavior of slime mould to search for better solutions

The Slime Mould Algorithm (SMA) addresses this problem by mimicking the food-searching, adaptive movement, and network-forming behavior of slime mould. It uses these biological behaviors to balance exploration and exploitation and search for an optimal solution.



OPTIMIZATION METHODS

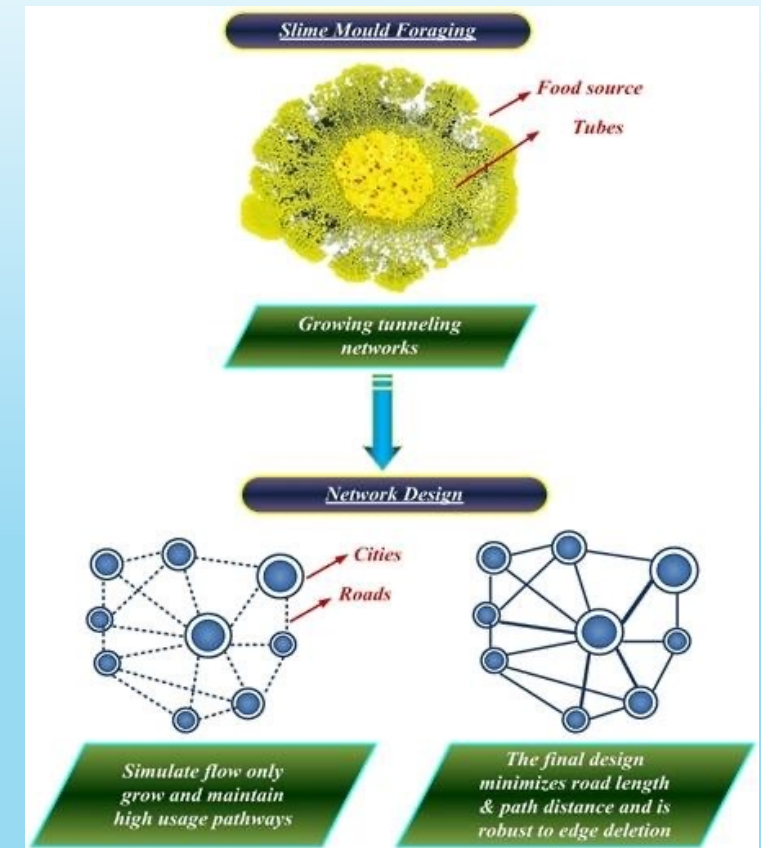
Data gathered through traditional condition method is often partisan, while bio inspired algorithms are an understudied and unbiased solution.

Traditional algorithm for solving optimization require immense computations and intuition

These optimization problems can be difficult because the search space may be complex, nonlinear, and contain many local optimum. solutions.

Traditional optimization methods may become slow or get trapped in a poor solution.

Therefore, there is a need for an efficient optimization technique that can explore different possible solutions and identify a near-optimal solution.



APPLICATIONS

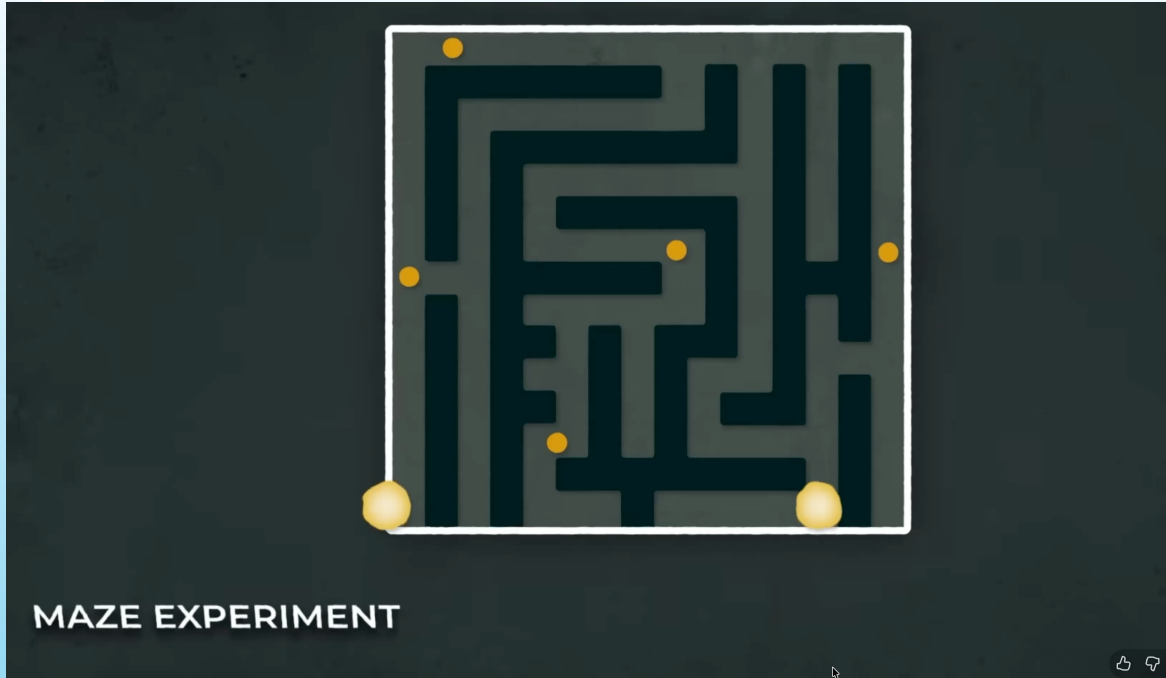
Slime Mould Optimization here majorly focuses on the intelligence, it can make choices that are well thought out which is not expected from such a small organism given that it does not even have brain but a very complex cellular network.

Before the 2000s, scientists discovered *Visarum plycepharum* (commonly know as slime mould) , *saw the way it moved* and classified it as a fungus.

Later, just around the year 2000 when researchers in Japan tested the *Visarum in a form of a maze, they got to know about its unique problem solving abilities.*

Lets learn how they researched :3

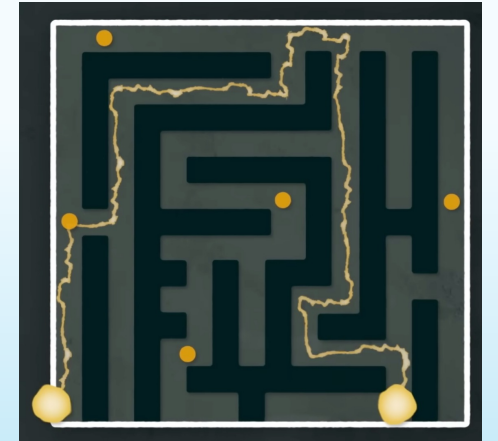
Toshiyuki Nakagaki's Maze Experiment



Source: [Real Science YT](#)



Possibility 1



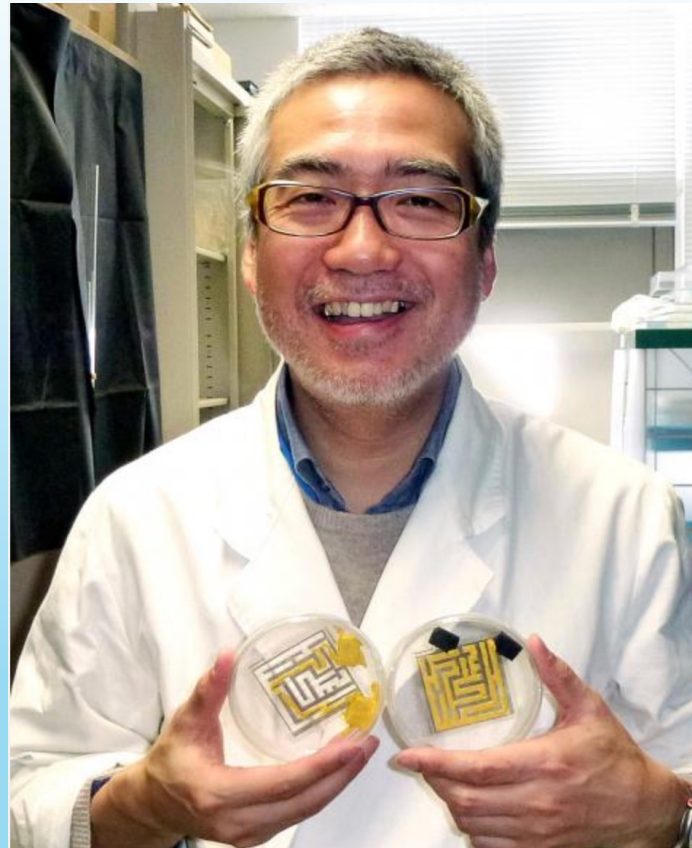
Possibility 2



Possibility 3



Possibility 4



(Toshiyuki Nakagaki, a professor at Future University Hakodate, shows amoeboid yellow slime mould on petri dishes in his laboratory in Hakodate, Japan, on March 10, 2011)

SOURCE: [TAIPEI TIMES](#)

EXTRA INFO ABOUT APPLICATIONS

Since Scientists got to know about the capabilities of *Visarum plycepharum*, they have been trying to find new applications as this mould is not only smart but they also automatically reconfigure their physical bodies to build the most efficient & shortest path just for the most nutrient dense networks possible.

Finding the shortest and the most optimised path between multiple points has many solutions which can be figured manually but it becomes difficult to select the one which is the most efficient and shortest which gets even more challenging and time consuming when more points and constraints are added, such type of problems often occur in the field of AI, Software, Engineering and Mathematics, using an living organism could help us to offer some great insights and solutions.

Slime Mould Algorithms [SMA] are basically optimization methods inspired by the foraging behaviour (behaviour in animals to catch food and survive) of this single celled organism.

TOKYO RAIL SYSTEM RECREATION

An Popular example of applying Slime Mould Optimisation in real life is when researchers tested the efficiency of *Visarum* trying to to see *how it maps out the Tokyo Rail System*.

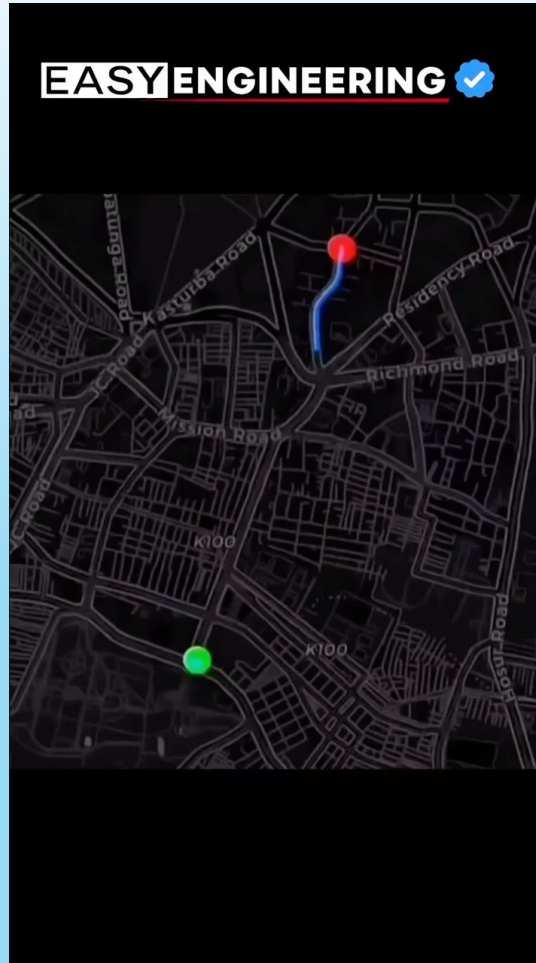
The main idea was arranging food source/oat flakes at all stations and keeping the Visarum in the centre of all the rail lines, the result of the experiment was that the Visarum had created a map which connected all the stations which was almost similar to the actual map of the railways then, this experiment was also led by Toshiyuki Nakagaki alongside Atsushi Tero (Researcher and Biologist) in partnership with University of Oxford and the University of the West of England.

Now, lets see an recreation video !



Source: [BBC Earth Science YT](#)

MAP ROUTING SYSTEMS USING SIMILAR ALGORITHMS



Disclaimer: This is not based on slime mould optimisation

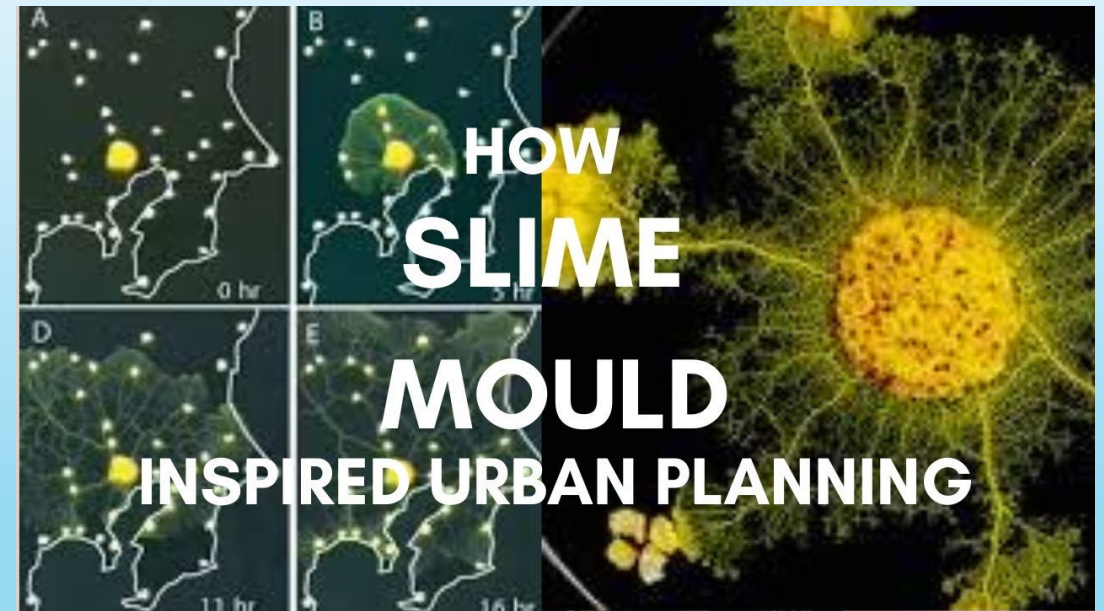
PROS

Pro 1: Finds Efficient Road Networks

The Slime Mould Algorithm helps urban planners design efficient road networks by searching through thousands of possible road layouts and selecting one of the best solutions. It naturally finds shorter and more connected routes, which reduces the distance people need to travel.

Pro 2: Reduces Traffic Congestion

Traffic congestion is one of the biggest challenges in modern cities. The Slime Mould Algorithm helps reduce congestion by optimizing traffic signal timings, road layouts, and vehicle movement. It distributes traffic more evenly across different roads, preventing overcrowding on major routes.



PROS

Pro 3: Finds Safe Navigation Paths

Space exploration missions often involve autonomous robots and rovers travelling across unknown and dangerous terrain. The Slime Mould Algorithm helps these vehicles find safe routes by avoiding obstacles such as rocks, craters, steep slopes, and rough terrain.

Pro 4: Saves Battery Power and Energy

Energy is one of the most valuable resources during space missions because spacecraft and rovers operate on limited battery power or solar energy. The Slime Mould Algorithm help conserve this energy by selecting shorter and more efficient travel routes. It reduces unnecessary movement and avoids difficult terrain that requires more power to cross.

CONS

Con 1: Does Not Always Find the Best Solution

Although the Slime Mould Algorithm usually finds very good solutions, it cannot guarantee that the solution is always the absolute best one. It is a heuristic optimization algorithm, meaning it searches intelligently rather than checking every possible solution.

- **Con 2: Requires High Computational Power**

The Slime Mould Algorithm must evaluate many possible solutions before selecting the best one. For large cities with thousands of roads, buildings, and traffic intersections, this process requires considerable computing power and memory.

CONS

Con 3: May Get Stuck in a Local Optimum

One limitation of the Slime Mould Algorithm is that it may sometimes become trapped in a local optimum. This means the algorithm finds a good solution but fails to discover an even better solution elsewhere in the search space. As a result, the selected route may not be the shortest or most energy-efficient possible.

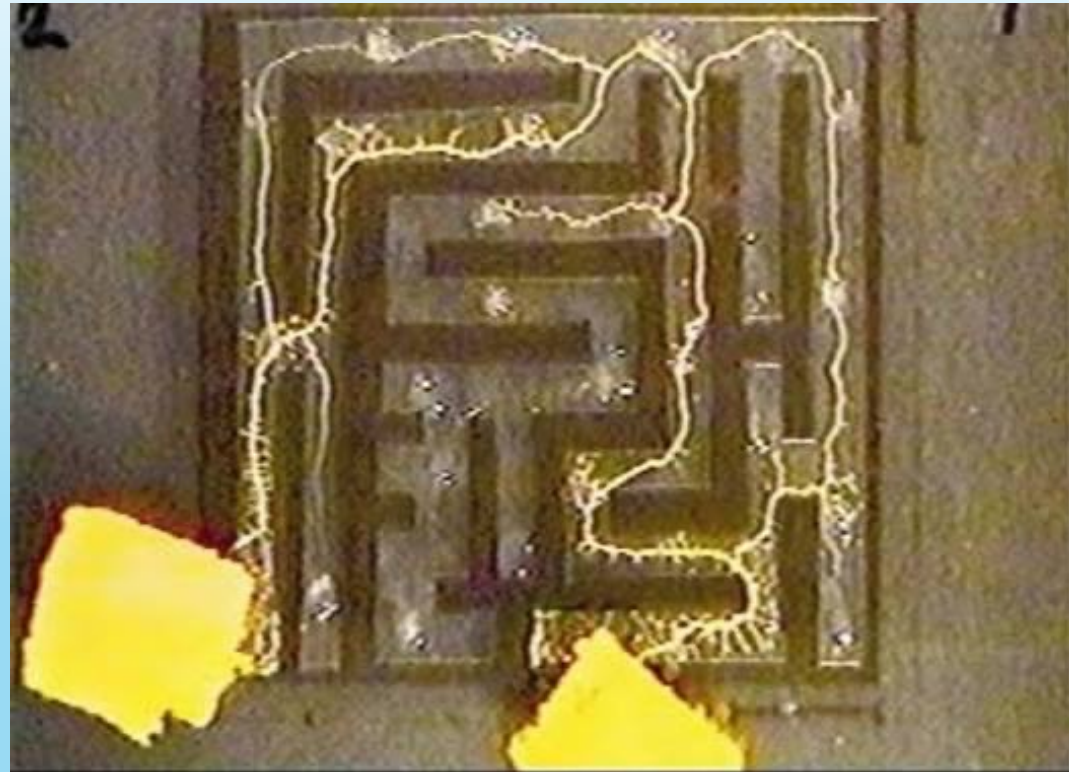
Con 4: Depends on Accurate Sensor Data

The effectiveness of the Slime Mould Algorithm depends heavily on the quality of information provided by the rover's sensors. Cameras, laser scanners, and other instruments must accurately detect rocks, craters, and other obstacles. If the sensor data is incorrect due to dust, poor lighting, or equipment failure, the algorithm may choose an unsafe or inefficient path.

RESEARCH TIMELINE - TIMES WHEN SLIME MOULD WAS USED BY HUMANS

2000 – Maze Solving

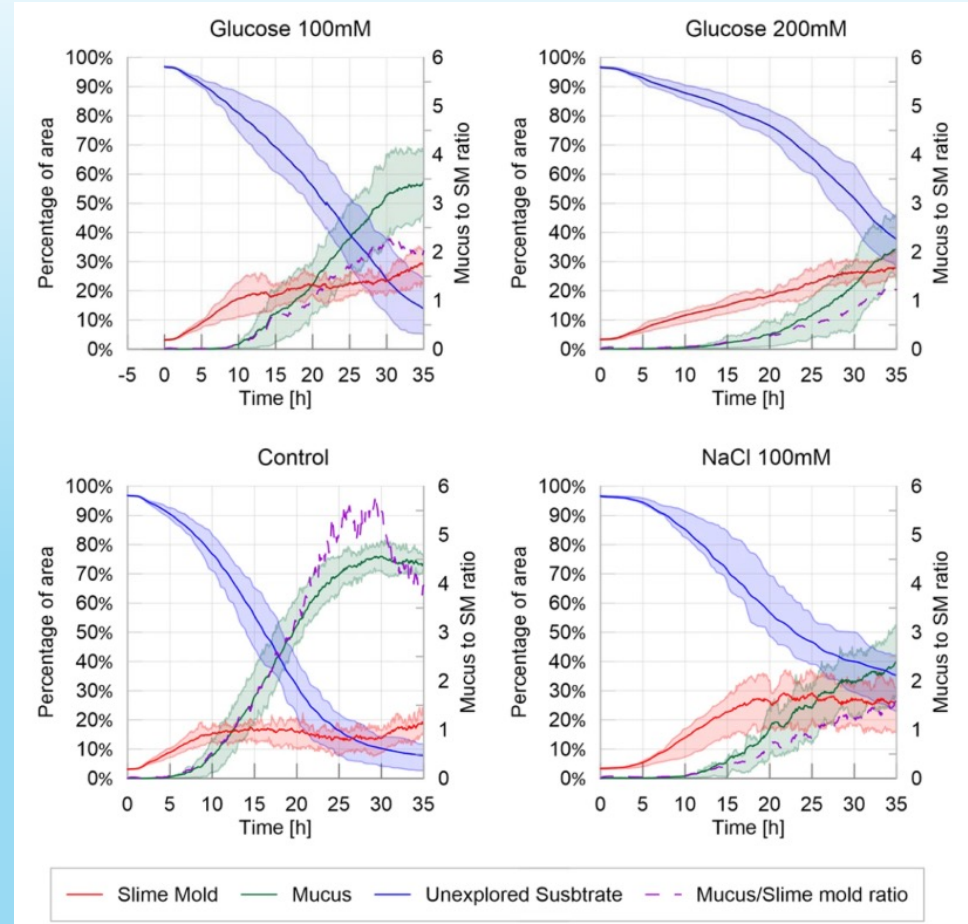
~[Maze-solving by an amoeboid organism](#)



2008 — ANTICIPATING PERIODIC EVENTS

In order to study the influence of the environment on slime mold expansion rate, we analyzed the areas covered by slime mold, unexplored substrate and mucus over time

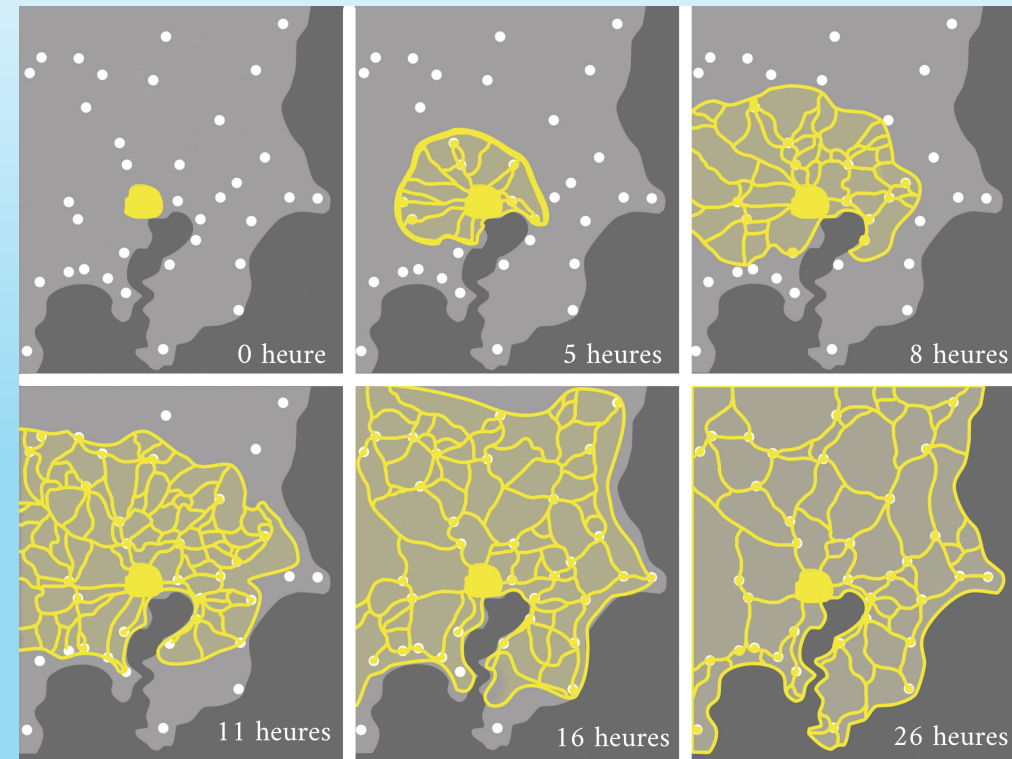
~[Figure 1 | Scientific Reports | Scientific Reports](#)



2010 — TOKYO RAIL NETWORK SIMULATION

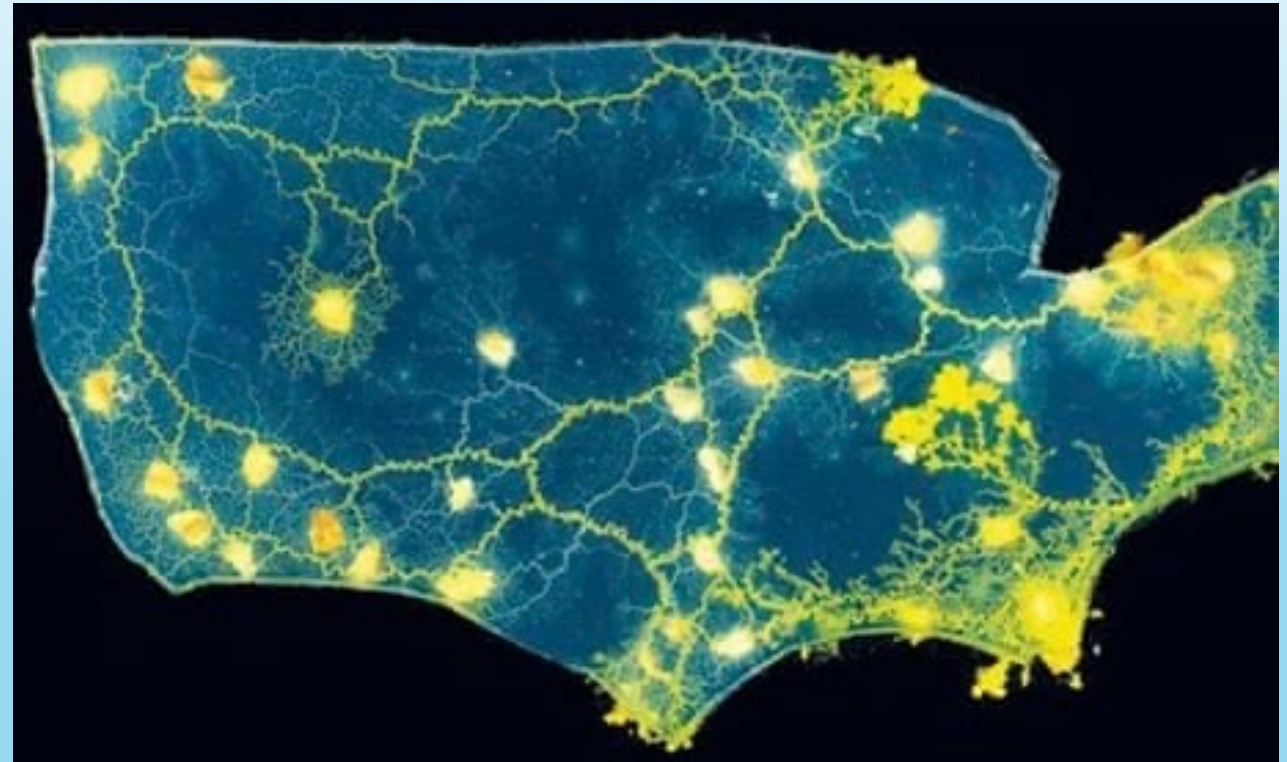
Physarum polycephalum network grown in a period of 26 hours (6 stages shown) to simulate greater Tokyo's rail network

~[Slime mold - Wikipedia](#)



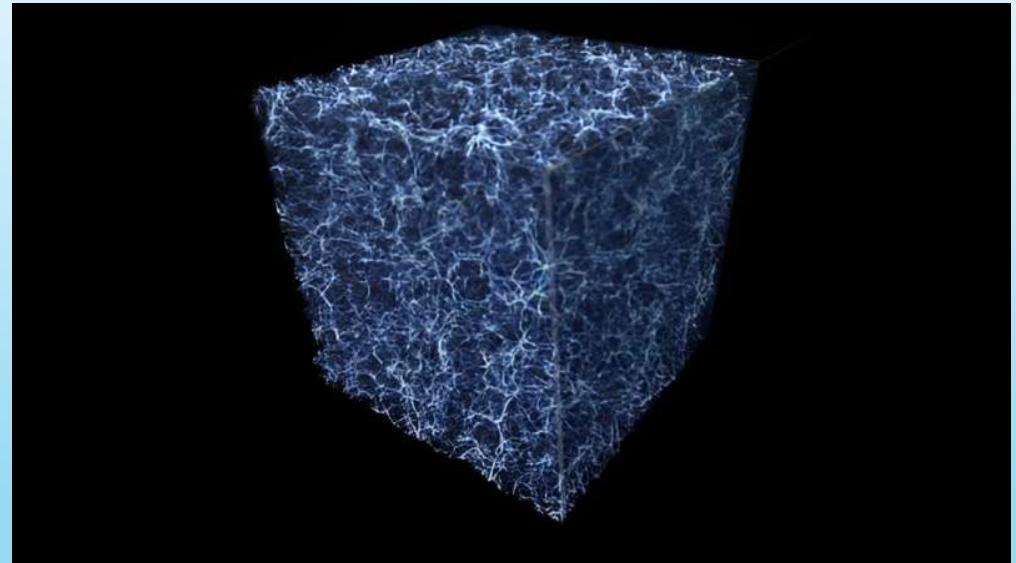
2010–2014 MOTORWAY NETWORK SIMULATIONS

Slime mould was used to approximate motorway networks in **14 countries** (Australia, Africa, Belgium, Brazil, Canada, China, Germany, Iberia, Italy, Malaysia, Mexico, Netherlands, UK, US). Example of the USA, Researchers use organism's search for food to work out efficient transport routes, divert around floods and even imitate rush hour



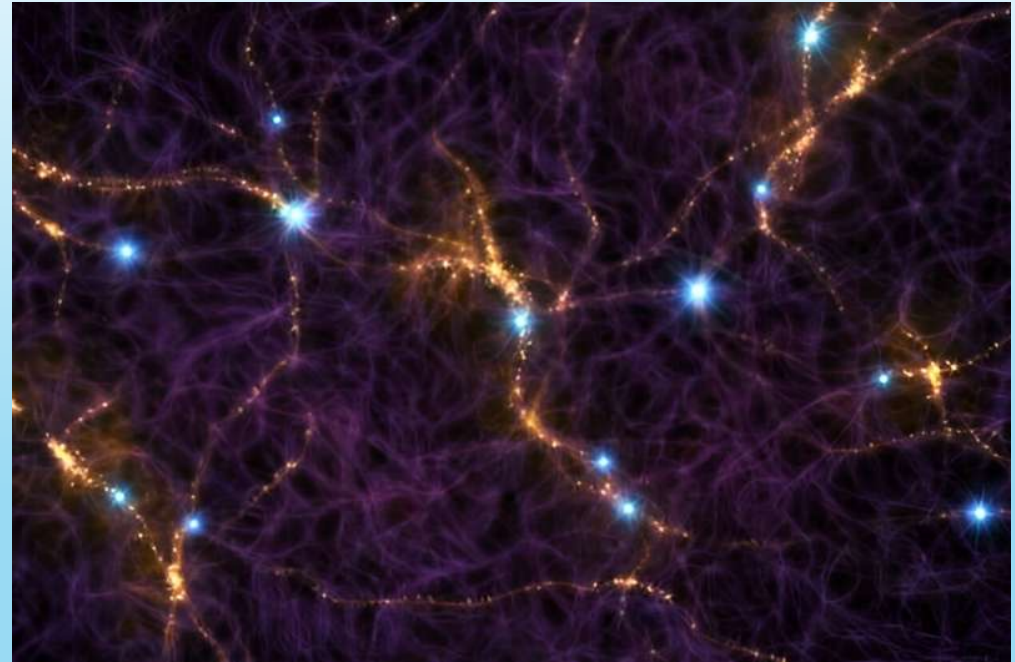
2020 — DARK MATTER MAPPING

Astronomers used slime mould simulations to model the filamentary structure of galaxies. The organism's vein networks resembled cosmic web structures, helping refine dark matter distribution models



SLIME MOULDS AND THE COSMIC WEB

The **large-scale structure of the universe** is a network of galaxies linked by vast filaments of gas, resembling the vein-like networks slime moulds create when foraging



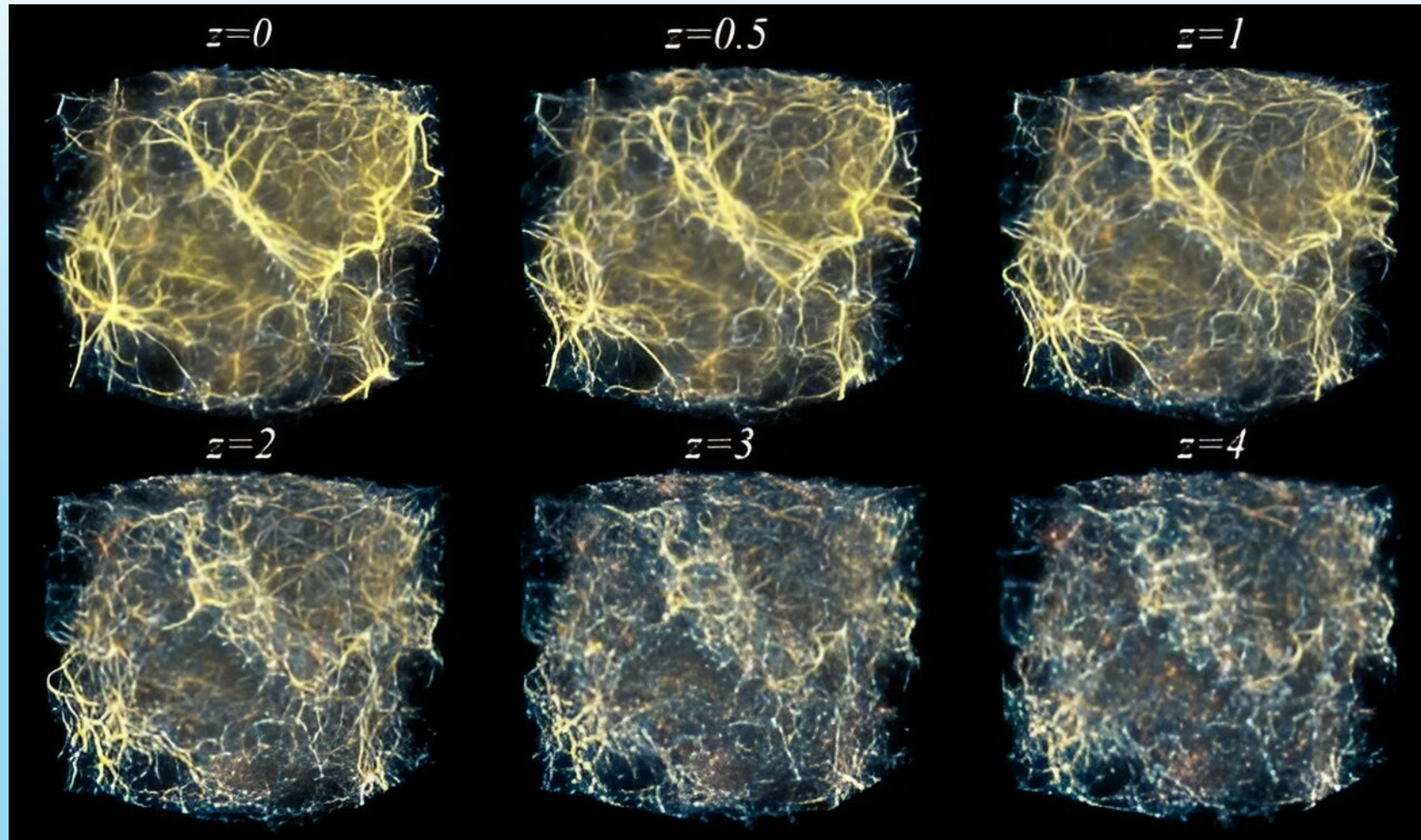
Researchers used a slime mould-inspired model, the **Monte Carlo Physarum Machine (MCPM)**, to reconstruct these cosmic filaments more accurately than traditional methods

Just as slime mould connects food sources with efficient tubular networks, MCPM connected galaxies and clusters, tracing the underlying **dark matter distribution** with natural, curved filament shapes .

Findings show that filaments can **supply gas to galaxies in the early universe**, fueling rapid growth, but later, thicker filaments **quench star formation** by restricting gas supply over the last 10 billion years .

This slime mould analogy helps astrophysicists understand how **environmental structures** influence galaxy evolution, complementing internal processes like supernovae and black hole activity .

SLIME MOULD BEING USED IN SPACE MAPPING



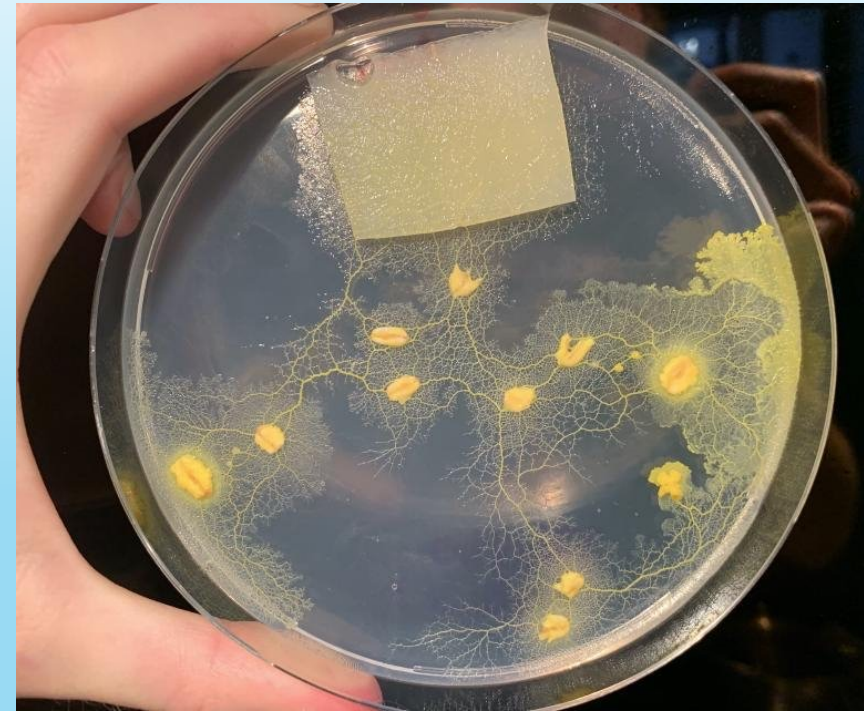
INTRODUCTION - STATISTICAL DATA

Physarum slime mould connecting food sources that were arranged in a pattern to resemble population centers in Hokkaido

Their aim was to answer a simple question:

"If slime mould had to build Britain's motorway system, what would it look like?"

The results were surprising because the network created by the slime mould was very similar to the UK's existing motorway network. This study became one of the most famous examples of **bio-inspired computing** and inspired later research into the Slime Mould Algorithm.



Optimizing transport networks using Physarum slime-mold

WHAT IS PHYSARUM POLYCEPHALUM?

Physarum polycephalum is not a fungus or a plant. It is a **single-celled amoeboid organism** that can grow to cover a large area.

Despite having **no brain or nervous system**, it can:

- Find the shortest path between food sources.

- Avoid obstacles.

- Repair damaged networks.

- Build efficient transport systems.



OBJECTIVE OF THE RESEARCH

The primary objective of this research was to investigate whether the natural behaviour of the slime mould *Physarum polycephalum* could be used to solve complex transportation network problems similar to those faced by civil engineers.

The researchers wanted to determine whether a simple organism with **no brain, no nervous system, and no central control** could naturally create an efficient transport network connecting major cities in the United Kingdom. They aimed to compare the biological network created by the slime mould with the existing UK motorway network and evaluate how closely the two matched.



HOW WAS THE EXPERIMENT CONDUCTED?

The researchers designed a laboratory experiment to test whether **Physarumpolycephalum** (slime mould) could naturally create an efficient transport network similar to the UK's motorway system. Instead of using computersoftware, they recreated the map of Great Britain inside a **Petri dish** and allowed the organism to make its own decisions. The entire experiment wascarefully controlled so that the slime mould's growth could be observed andcompared with the real UK road network

The 10 urban areas represented in the experiment were:

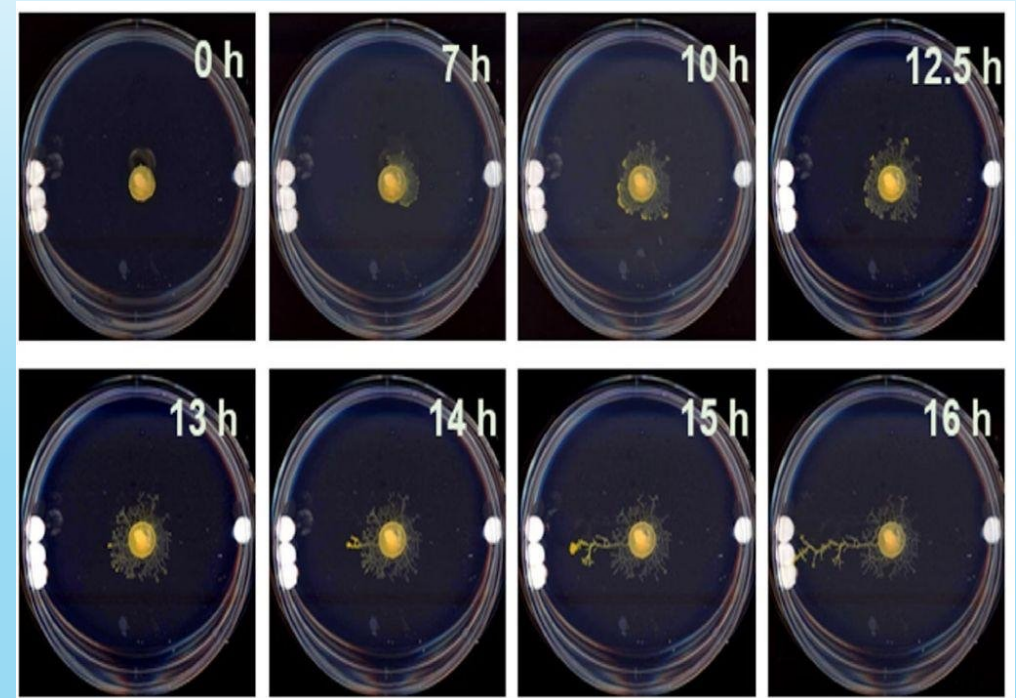
London

Birmingham

Bristol

Sheffield

Leeds



WHAT HAPPENED DURING THE EXPERIMENT?

During the experiment, the researchers did **not control or guide** the slime mould in any way. Once the organism was placed on the oat flake representing **Greater London**, it was allowed to grow naturally. Every movement and every connection it made was based entirely on its biological behaviour. The researchers simply observed and recorded how the network developed over time. This made the experiment a true demonstration of **self-organization**, where a living organism solved a complex problem without any external instructions.

RESULTS OF THE EXPERIMENT

The results of the experiment were remarkable because the slime mould successfully created a transport network that closely resembled the existing motorway network of Great Britain. Even though the organism had **no brain, no memory, and no map**, it naturally connected the major cities using efficient routes. This demonstrated that simple biological processes can solve complex network optimization problems that are usually handled by engineers and computer algorithms. The researchers concluded that the slime mould's behaviour follows principles of **efficiency, connectivity, and adaptability**, making it a valuable model for designing transportation and communication networks

IMPORTANCE OF THE RESEARCH

The UK slime mould experiment is considered one of the most important studies in **bio inspired computing** because it showed that a very simple living organism can solve a complex transportation problem without having a brain, memory, or central control. Before this research, scientists believed that designing efficient transport networks required advanced mathematical calculations and human planning.

However, the experiment demonstrated that natural systems can also produce highly efficient solutions using simple biological rules. This changed the way researchers think about solving optimization problems and inspired many new areas of scientific research.



STATISTICAL HIGHLIGHTS

The UK slime mould experiment is often described using statistics to show how closely the biological network resembled the real motorway network. While the study was primarily **qualitative** (observing patterns and comparing networks), it also reported several measurable facts that support its conclusions.

Research Fact	Value 
Research year	2009
Researchers	Andrew Adamatzky & Jeff Jones
University	University of the West of England (UWE), Bristol
Organism used	Physarum polycephalum
Cities represented	10 major UK urban areas
Starting location	Greater London
Major outcome	Network closely resembled the UK's motorway system
Main difference	Preferred England–Scotland link via Newcastle instead of the M6/M74 corridor
Additional experiment	Network successfully reorganized after simulated disruptions

SUMMARY

1) Problem Statement:

- a) Many real-world optimization problems have a very large search space.
- b) Traditional methods can get trapped in local optima.
- c) Some algorithms may have slow convergence or poor exploration.
- d) SMA aims to find an optimal or near-optimal solution efficiently.

2) Biological Phenomenon:

- a) Slime mould searches for food by spreading its network of tubes.
- b) It adapts its pathways according to the quality of food sources.
- c) Strong food sources attract stronger connections.
- d) SMA converts this adaptive behavior into mathematical optimization rules.

3) Applications:

- a) Engineering design optimization
- b) Scheduling and resource allocation
- c) Image processing
- d) Path planning
- e) Feature selection
- f) Machine learning optimization

4) Research Timeline:

- a) 2020: SMA introduced by Li et al.
- b) 2021–2022: SMA variants and improvements developed.
- c) 2023–2024: Increasing applications in engineering, machine learning and optimization.
- d) 2025 onward: Research focuses on hybrid, adaptive and application-specific versions.

5) Statistical Data:

- a) SMA is commonly evaluated using benchmark optimization functions.
- b) Results across studies generally show that SMA can be competitive, although performance depends on the problem and experimental setup.

6) Future Possibilities:

- a) Hybrid SMA with AI and machine learning
- b) Adaptive parameter control
- c) Multi-objective optimization
- d) Real-time optimization
- e) Applications in robotics and autonomous systems
- f) Energy and transportation optimization

FUTURE POSSIBILITIES

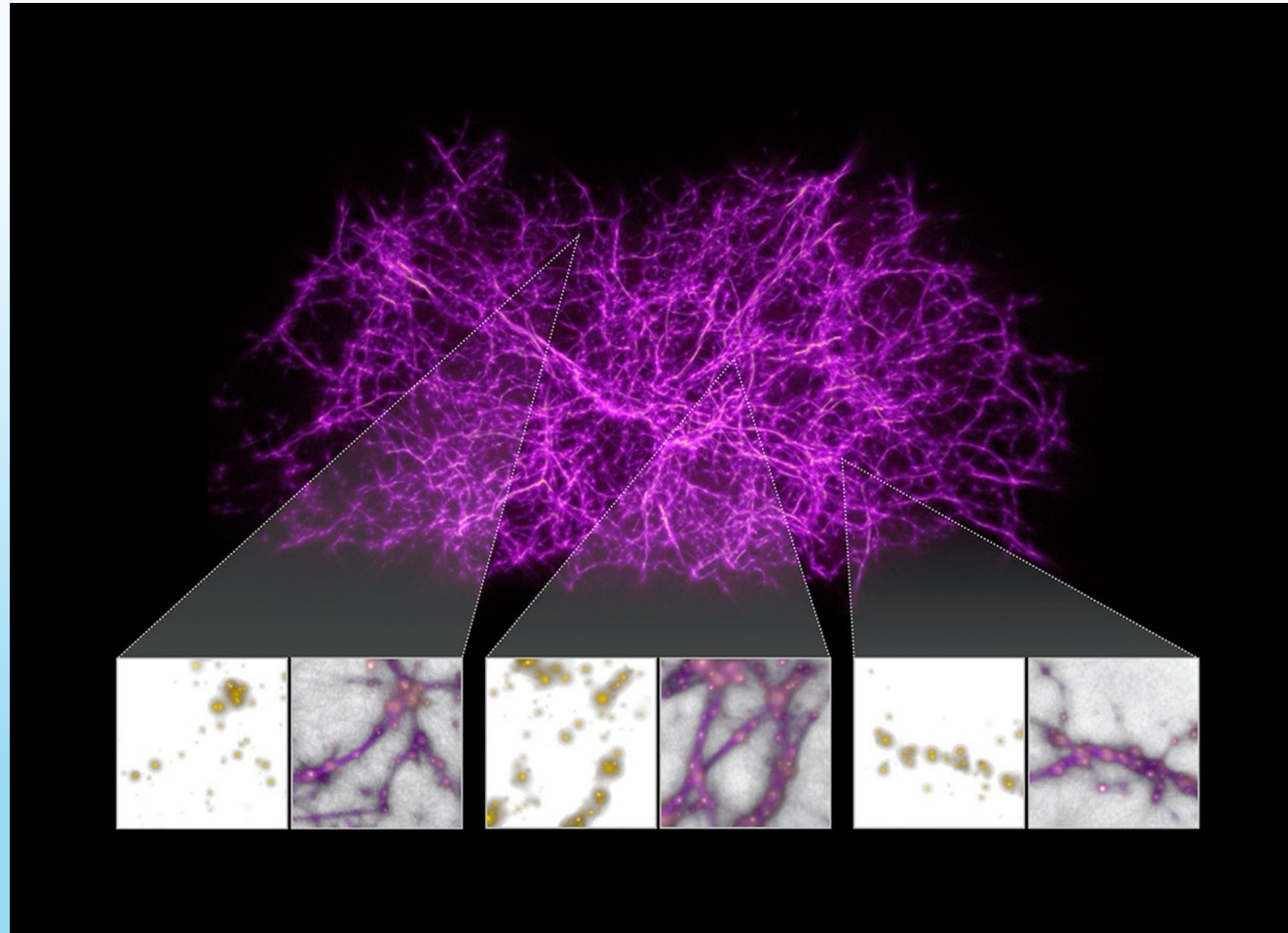
Slime Mould Optimization (SMO) has good potential for future urban infrastructure planning and autonomous space exploration by doing biomimicry of this organism's smart decision making and intelligence.

In urban planning, future models could move from testing simulations which take up a lot of time labour and money for say any upcoming plans or cities, we could use slime mould to turn the plans into real, efficient and optimised transit and networks capable of routing around constraints like structural damage or congestion as this mould can also adapt to such situations.

For Space exploration, a group of some astronomers from UC Santa Cruz along and computational from NASA tried to join the invisible filaments like distant galaxies to connect our universe and trace the map of invisible cosmic web of dark matter and gas connecting galaxies.



Overlapped
Mumbai Metro Map and
Slime Mould Network



Cosmic Web and Slime Mold Photo ([NASA](#))

Video of How a Living Computer was Created with Slime Mould



Source: [Seeker YT](#)



THANK YOU